Table 1: Per-benchmark results for the 100B token MixtureVitae mixture composition ablation on the standard language understanding suite. Removing the instruction and reasoning subset creates a persistent language gap that Phase 1 does not close, whereas removing the curated web subset leaves language performance close to the full mixture across stages and both context lengths. Each value is one run, printed with one decimal, and Avg is the mean of the eleven tasks.

Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
MV-noinstruct	32.1	60.6	58.3	18.3	66.0	45.7	45.6	25.1	32.6	68.2	54.6	46.1
→SFT	32.1	60.7	57.2	19.3	66.0	46.5	46.6	25.1	32.8	67.9	55.1	46.3
→SFT →DPO	32.3	60.8	57.4	19.2	66.0	46.5	46.6	25.2	32.8	67.8	55.1	46.3
MV-noinstruct-16k	32.7	60.3	63.3	19.5	69.0	45.6	47.0	24.8	32.6	66.8	54.7	46.9
→SFT	31.3	59.8	60.7	18.3	67.0	46.3	46.7	24.9	32.8	66.9	55.7	46.4
→SFT →DPO	31.3	59.8	60.7	17.9	67.0	46.5	46.9	24.9	32.4	66.9	55.9	46.4
MV-noweb	34.5	64.7	66.6	34.6	67.0	46.2	38.7	31.4	33.0	67.5	56.0	49.1
→SFT	32.8	63.0	64.8	35.2	69.0	46.7	40.6	33.0	31.8	67.2	55.2	49.0
→SFT →DPO	32.8	63.0	64.8	35.5	69.0	46.8	40.7	33.0	32.0	67.2	55.0	49.1
MV-noweb-16k	35.6	65.2	69.8	36.6	70.0	45.4	40.5	32.1	32.4	67.1	55.8	50.0
→SFT	32.8	63.3	66.4	34.2	69.0	46.4	41.0	33.1	32.8	67.0	56.2	49.3
→SFT →DPO	32.9	63.5	66.6	34.2	69.0	46.5	41.2	33.1	33.2	67.0	56.4	49.4
MV	34.4	64.6	70.9	31.4	67.0	48.5	43.8	32.5	33.8	68.4	57.7	50.3
→SFT	33.9	63.8	70.5	31.9	69.0	49.3	45.0	33.0	33.6	68.7	57.7	50.6
→SFT →DPO	33.9	64.0	70.7	32.0	70.0	49.4	45.0	33.1	33.4	68.7	57.5	50.7
MV-16k	34.7	65.0	67.7	29.9	67.0	47.1	44.6	32.6	32.8	67.9	57.1	49.7
→SFT	33.4	64.2	69.2	29.2	68.0	48.8	45.0	32.4	32.6	68.3	58.7	50.0
→SFT →DPO	33.4	64.3	69.4	29.6	68.0	49.0	45.2	32.4	32.8	68.5	58.9	50.1

Table 2: Full language suite results for the main 1.7B comparison. Phase 1 Tulu3 SFT/DPO leaves language scores largely unchanged, with MixtureVitae matching FineWeb-Edu, trailing Nemotron, and remaining below the more heavily pretrained external baselines. Each value is one run, printed with one decimal, and Avg is the mean of the eleven tasks. The Nemotron →SFT and →SFT →DPO rows are the 16K checkpoints.

Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
Comma0.1-300BT	36.2	63.1	63.2	20.4	70.0	53.5	54.5	26.5	32.6	71.2	59.3	50.0
→SFT	36.5	64.6	64.1	20.3	74.0	54.0	57.0	27.2	33.2	71.9	60.9	51.3
→SFT →DPO	36.7	64.7	64.7	20.5	74.0	54.2	57.3	27.2	33.2	71.9	60.9	51.4
MV	40.3	70.9	73.5	51.4	71.0	54.5	48.1	41.9	34.2	70.4	57.8	55.8
→SFT	38.8	70.2	73.2	49.6	70.0	55.7	49.6	42.7	34.4	70.0	58.2	55.7
→SFT →DPO	38.8	70.2	73.2	50.0	71.0	55.8	49.8	42.7	34.4	70.0	58.2	55.8
MV-16k	41.6	71.4	72.6	48.8	67.0	53.1	49.4	41.1	31.8	69.7	57.9	55.0
→SFT	39.2	70.2	72.9	50.4	69.0	54.9	50.0	41.9	34.2	69.9	57.5	55.5
→SFT →DPO	39.2	70.5	72.9	50.3	69.0	55.0	50.2	42.0	34.4	69.8	57.4	55.5
Qwen2.5-1.5B	53.8	80.5	78.4	75.8	83.0	68.0	63.0	61.0	40.8	77.1	63.4	67.7
→SFT	51.2	79.0	69.0	76.6	82.0	68.0	60.7	60.5	39.4	77.0	62.4	66.0
→SFT →DPO	51.6	79.1	69.3	76.6	82.0	68.4	60.6	60.6	39.2	77.0	62.4	66.1
Qwen3-1.7B	54.0	82.2	82.6	75.8	82.0	67.1	63.9	62.7	39.0	76.6	64.4	68.2
→SFT	53.0	80.7	82.6	76.5	80.0	67.7	61.9	62.9	38.6	76.1	63.9	67.6
→SFT →DPO	53.5	80.9	82.7	76.4	81.0	68.3	61.9	63.0	39.0	76.3	63.6	67.9
SmolLM2-1.7B	53.6	79.6	74.3	60.5	83.0	73.1	67.7	50.4	44.6	78.9	66.1	66.5
→SFT	52.4	80.5	74.9	60.2	85.0	73.8	69.7	49.2	43.8	78.7	67.1	66.8
→SFT →DPO	53.0	80.7	74.8	60.5	85.0	74.1	69.8	49.3	43.6	78.7	67.0	67.0
FineWeb-Edu-300BT	48.5	76.9	64.9	21.1	82.0	67.1	55.7	26.1	42.4	77.1	62.4	56.7
→SFT	48.1	78.5	65.9	19.7	83.0	67.6	59.2	27.0	41.8	77.0	63.2	57.4
→SFT →DPO	48.4	78.5	65.9	19.7	83.0	67.8	59.3	26.9	42.0	77.1	63.4	57.5
Nemotron	52.4	80.9	77.4	48.6	85.0	71.5	58.0	46.4	42.0	78.7	65.0	64.2
→SFT	48.0	78.7	74.3	46.2	82.0	70.5	58.5	43.0	42.8	78.2	64.6	62.4
→SFT →DPO	48.3	78.9	74.4	45.9	82.0	70.7	58.8	43.2	42.8	78.2	64.6	62.5

Table 3: Language suite of 300B MixtureVitae checkpoints trained without and with the historical 13-gram filtering of their data, at 4K and 16K context, before and after Phase 1 Tulu3 SFT/DPO. Each value is one run, printed with one decimal, and Avg is the mean of the eleven tasks.

Base	Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
Without filtering	MV	39.6	71.3	75.0	48.7	73.0	53.9	48.2	38.0	35.2	69.6	58.3	55.5
	→SFT	39.0	70.4	74.6	46.8	71.0	54.9	49.8	39.8	35.0	70.1	58.3	55.4
	→SFT →DPO	39.1	70.5	74.7	47.0	71.0	55.1	49.9	39.9	35.0	70.1	58.3	55.5
	MV-16k	39.0	71.0	74.6	50.1	72.0	54.3	48.7	37.7	35.6	70.8	57.7	55.6
	→SFT	38.8	70.5	74.4	48.4	72.0	54.8	49.6	38.0	34.6	70.5	58.7	55.5
	→SFT →DPO	39.1	70.7	74.5	48.2	72.0	55.0	49.9	38.1	35.0	70.5	58.7	55.6
With filtering	MV	40.3	70.9	73.5	51.4	71.0	54.5	48.1	41.9	34.2	70.4	57.8	55.8
	→SFT	38.8	70.2	73.2	49.6	70.0	55.7	49.6	42.7	34.4	70.0	58.2	55.7
	→SFT →DPO	38.8	70.2	73.2	50.0	71.0	55.8	49.8	42.7	34.4	70.0	58.2	55.8
	MV-16k	41.6	71.4	72.6	48.8	67.0	53.1	49.4	41.1	31.8	69.7	57.9	55.0
	→SFT	39.2	70.2	72.9	50.4	69.0	54.9	50.0	41.9	34.2	69.9	57.5	55.5
	→SFT →DPO	39.2	70.5	72.9	50.3	69.0	55.0	50.2	42.0	34.4	69.8	57.4	55.5

Table 4: Language suite results for the smaller scale comparison between MV (0.4B) and Baguettotron. MV retains a small but consistent advantage across both context lengths and post-training stages, showing that its broader general-purpose edge at small scale is not limited to reasoning tasks. Each value is one run, printed with one decimal, and Avg is the mean of the eleven tasks.

Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
Baguettotron	32.8	58.5	60.7	19.6	56.0	36.3	29.8	26.5	30.8	60.9	53.3	42.3
→SFT	30.5	57.5	61.1	21.5	61.0	36.3	33.0	26.2	31.6	60.9	54.6	43.1
→SFT →DPO	30.7	57.5	61.6	21.0	61.0	36.4	33.2	26.2	31.6	60.8	54.4	43.1
Baguettotron 16k	31.6	57.8	57.4	20.6	60.0	35.8	33.7	26.5	31.0	61.5	55.2	42.8
→SFT	30.5	57.2	63.6	19.7	61.0	36.3	33.0	25.8	31.4	60.9	54.4	43.1
→SFT →DPO	30.5	57.2	63.7	19.5	61.0	36.3	33.4	25.8	31.6	61.0	54.3	43.1
MV (0.4B)	31.7	60.6	59.8	25.6	63.0	42.0	37.6	28.4	30.8	65.5	53.1	45.3
→SFT	29.0	58.7	50.1	24.9	63.0	42.3	39.7	28.5	30.2	65.4	53.4	44.1
→SFT →DPO	29.2	59.0	50.2	25.6	63.0	42.4	39.8	28.7	30.2	65.5	53.9	44.3
MV-16k (0.4B)	32.0	61.1	57.9	25.8	63.0	40.6	40.6	28.4	31.8	64.7	53.6	45.4
→SFT	28.2	58.7	53.4	25.1	64.0	41.7	40.4	28.6	30.6	64.9	55.1	44.6
→SFT →DPO	28.3	58.8	53.5	25.1	63.0	41.7	40.7	28.6	30.6	64.6	54.9	44.5

Table 5: Language suite results after Phase 2 OpenThoughts3 30K post-training for the models evaluated in the OT3 experiments. MV-16K's general language understanding stays near its base level, after OT3 from the SFT checkpoint, after the full pipeline and after direct OT3 post-training from the 16K extended checkpoint alike. Each value is one run, printed with one decimal, and Avg is the mean of the eleven tasks.

Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
MV-16k												
→SFT →OT30K	39.3	70.3	73.5	49.6	71.0	55.2	49.0	41.3	34.4	70.1	58.6	55.7
→SFT →DPO →OT30K	39.7	70.0	73.3	50.4	71.0	55.5	48.9	41.3	33.8	70.4	58.9	55.7
→OT30K (**without Tulu3**)	42.5	70.9	74.5	49.1	69.0	54.1	48.7	41.3	33.2	70.1	58.6	55.6
Comma-16k												
→SFT →OT30K	38.0	66.5	66.1	21.6	71.0	53.8	53.7	27.6	33.8	71.4	60.9	51.3
→SFT →DPO →OT30K	36.0	65.8	65.1	22.0	73.0	51.1	49.3	27.5	33.4	70.1	58.9	50.2
FineWeb-Edu-16k												
→SFT →OT30K	47.4	78.5	69.5	21.4	83.0	66.7	54.0	26.8	40.8	77.1	63.2	57.1
Nemotron-16k												
→SFT →OT30K	48.0	79.0	75.4	45.7	81.0	70.0	56.8	41.0	42.0	77.8	64.2	61.9
→SFT →DPO →OT30K	48.5	79.2	75.4	46.8	81.0	70.0	57.2	41.3	42.4	77.9	64.6	62.2
Qwen2.5-1.5B												
→SFT →OT30K	50.8	78.1	76.7	72.8	84.0	67.3	46.3	58.8	41.2	76.9	63.1	65.1
→SFT →DPO →OT30K	50.9	78.0	76.9	72.6	84.0	67.5	46.2	58.8	40.6	77.0	63.3	65.1
Qwen3-1.7B												
→SFT →OT30K	49.1	76.1	74.5	65.0	81.0	59.3	42.8	53.0	38.6	74.0	60.5	61.3
→SFT →DPO →OT30K	49.0	75.8	74.8	64.5	81.0	59.5	43.0	53.1	38.6	73.6	60.5	61.2
SmolLM2-1.7B-16k												
→SFT →OT30K	52.6	80.8	70.4	58.6	88.0	70.9	64.1	44.4	43.0	77.5	66.1	65.1
SmolLM2-1.7B-Instruct-16k												
→OT30K	51.4	79.4	75.9	62.2	83.0	71.1	58.6	47.4	44.4	77.9	65.7	65.2

Table 6: Language understanding (open-sci suite) with vs. without prompt-token loss masking in Phase-1 post-training (1.7B MixtureVitae, decontaminated). Each value is one run, printed with one decimal, and Avg is the mean of the eleven tasks. The rows with masking are the `laion` masked checkpoints.

	Model	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
With masking	MV												
	→ SFT	38.7	70.1	74.6	51.8	70.0	54.9	50.1	42.5	34.6	70.2	58.5	56.0
	→ SFT → DPO	38.5	70.2	74.6	51.7	71.0	55.0	50.3	42.6	34.8	70.2	57.8	56.1
	MV-16k												
	→ SFT	39.3	70.5	74.9	50.9	68.0	53.6	50.5	42.1	34.4	70.0	57.6	55.6
	→ SFT → DPO	39.4	70.5	75.1	50.6	69.0	53.7	50.7	42.1	34.2	70.1	57.6	55.7
Without masking	MV												
	→ SFT	38.8	70.2	73.2	49.6	70.0	55.7	49.6	42.7	34.4	70.0	58.2	55.7
	→ SFT → DPO	38.8	70.2	73.2	50.0	71.0	55.8	49.8	42.7	34.4	70.0	58.2	55.8
	MV-16k												
	→ SFT	39.2	70.2	72.9	50.4	69.0	54.9	50.0	41.9	34.2	69.9	57.5	55.5
	→ SFT → DPO	39.2	70.5	72.9	50.3	69.0	55.0	50.2	42.0	34.4	69.8	57.4	55.5